

Standard Operating Procedure

Assembly Line Product Changeover

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1. Purpose

This procedure defines the mandatory steps for performing a product changeover on NEXUS assembly lines. The changeover procedure ensures complete removal of all components from the previous product before the next product is introduced, preventing cross-contamination and component mix-up between different NEXUS products.

2. Scope

Applicable to all NEXUS assembly lines (AL-001 through AL-005) at Marcy-l'Etoile when changing from one NEXUS product to another. This includes changeovers between any NEXUS product variants (CK MB, B2M, Troponin, D-Dimer, etc.).

WARNING: MINIMUM CHANGEOVER TIME: 30 MINUTES. Production of the new product MUST NOT begin until at least 30 minutes have elapsed since the last unit of the previous product exited the assembly line. This time is required to complete all 8 items on the clearance checklist. Any deviation from this minimum time must be documented as a non-conformity per SOP-QA-010.

3. References

- FMEA-CKMB-001: FMEA for NEXUS CK MB
- FMEA-B2M-001: FMEA for NEXUS B2M
- SOP-QA-010: Non-Conformity Reporting
- FORM-ASM-003: Assembly Line Clearance Checklist
- INV-2024-0067: Investigation - Assembly Line Mix-up (reference case)

4. Procedure

4.1 Pre-Changeover

- 4.1.1 Confirm that the last production lot of the current product has been completed and all units have exited the assembly line.
- 4.1.2 Record the completion time on FORM-ASM-003.
- 4.1.3 Start the changeover timer (electronic timer on line control panel).
- 4.1.4 Notify the area supervisor that changeover is beginning.

4.2 Clearance Checklist (8 Items)

Item	Step	Responsibility	Acceptance Criteria
1	Remove all previous product materials from the line	Operator	All trays, containers, and loose materials collected and returned to storage
2	Empty and clean feed conveyors	Operator	Conveyors run empty for 2 minutes, then wiped with IPA
3	Clear and verify SPR cone hopper is empty	Operator	Hopper visually inspected from top and bottom; confirm zero cones remaining
4	Replace reagent strip magazine with new product strips	Operator	Old strips removed and returned; new product strips loaded and barcode verified
5	Change label rolls to new product labels	Operator	Old rolls removed; new rolls loaded; first label printed and verified against spec
6	Verify barcode scanner is operational (NOT in bypass)	Operator	Scan test barcode; confirm scanner reads and validates correctly. If scanner fails, stop line
7	Perform first article inspection	Operator + QC	Assemble one test kit; verify all components match new product specification
8	Supervisor sign-off	Supervisor	Supervisor physically present, verifies all items 1-7 completed, signs FORM-ASM-003

WARNING: ALL 8 ITEMS must be completed and signed before production of the new product can begin. Item 3 (hopper clearance) and Item 6 (barcode scanner) are CRITICAL controls - failure to verify these items has resulted in product mix-ups (ref: INV-2024-0067).

4.3 Post-Changeover Verification

- 4.3.1 Verify that the electronic changeover timer shows >= 30 minutes before authorising production start.
- 4.3.2 Verify that the line control system shows the correct product reference for the new product.
- 4.3.3 Run first 10 units through the line and verify barcode scan results (all must show correct product).
- 4.3.4 If any unit fails barcode verification: STOP the line immediately and repeat the changeover procedure from step 4.2.1.

5. Product Identification Guide

Product	Reference	SPR Code	SPR Cap Colour	Strip Colour
NEXUS CK MB	30421	SPR-CKMB-001	Red cap	Grey
NEXUS B2M	30420	SPR-B2M-001	Blue cap	White
NEXUS Troponin	30450	SPR-TROP-001	Green cap	Yellow
NEXUS D-Dimer	30460	SPR-DDIM-001	Orange cap	Blue

6. Deviation Handling

- 6.1 If changeover cannot be completed within 30 minutes (e.g., equipment fault): document the delay on FORM-ASM-003 and notify supervisor.
- 6.2 If any checklist item cannot be satisfactorily completed: STOP and raise a non-conformity per SOP-QA-010 before proceeding.
- 6.3 If the barcode scanner is in bypass/fault: production MUST NOT start. Contact maintenance for immediate repair (priority: URGENT).
- 6.4 If a component mix-up is discovered after production has started: immediately stop the line, quarantine all units produced since changeover, and open NCR per SOP-QA-010.

7. Revision History

Rev	Date	Author	Changes
1	2023-01-15	L. Renard	Initial release
2	2024-09-01	L. Renard	Added electronic timer requirement, mandatory supervisor physical presence (items from CAP

8. Approval

Role	Name	Signature	Date
Author	J. Martin		
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